

DETECTION OF AI-GENERATED IMAGES USING FINE-TUNED CNNs WITH INTERPRETABILITY TECHNIQUES

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ABSTRACT

The detection of AI-generated images has become increasingly critical as generative AI models produce highly realistic synthetic images. Using the CIFAKE dataset, our fine-tuned CNN models achieved a test accuracy of 95.2% when applying cross-entropy as loss function and 94.7% when applying binary cross-entropy, which compares favorably against other state-of-the-art models. For instance, recent studies demonstrated accuracies of 92.98% [1], 96.31% [2] and 97.32% [3] using advanced CNN architectures. To enhance interpretability, GradCAM was applied to visualize the regions most influential to the model's decisions. This combination of high accuracy and explainability demonstrates the potential of CNNs in synthetic image detection tasks, providing tools for improving image authenticity verification.

Index Terms— AI-generated images, CNN, GradCam

1. INTRODUCTION

The rapid advancement in generative AI has led to synthetic images indistinguishable from real images, posing challenges in applications such as digital forensics, media authenticity, and privacy protection. The ability to differentiate between real and AI-generated images is crucial in establishing the authenticity of visual content. This project will develop a Convolutional Neural Network (CNN) to classify synthetic vs. authentic images, using the CIFAKE dataset as a benchmark. State-of-the-art CNN models, as highlighted in prior research, have achieved impressive results, with accuracies of 92.98% [1], 96.31% [2] and 97.32% [3].

Deep learning models, especially CNNs, have shown remarkable success in image classification. However, distinguishing synthetic images requires nuanced detection of fine-grained artifacts that only advanced CNN architectures and interpretable methods can reveal. Explainable AI techniques, such as GradCAM saliency maps, enhance model transparency by visualizing image regions that influence decision-making. The CIFAKE dataset provides a balanced set of real and synthetic images, making it an ideal testbed for training and evaluating our CNN model [4].

This project aims to: Build a fine-tuned CNN for binary classification of real vs. synthetic images, enhance model transparency using GradCAM, evaluate the performance and

the interpretability of the model in order to contribute further to the field of synthetic image detection.

2. METHODS

2.1. Dataset and Preprocessing

The CIFAKE dataset contains 60,000 images from the CIFAR-10 dataset and 60,000 images generated by a latent diffusion model [1]. The dataset is balanced wrt. subclasses of the CIFAR-10 dataset. This was verified by extracting the subclass identifier from the image filenames using regular expressions to isolate numbers inside parentheses. Each subclass contains 12,000 RGB images with 32×32 pixels in .jpg format. When downloading the dataset [5], it is already split into a training and test set, where the images are labeled REAL or FAKE. To prevent overfitting, we have split the training data into a training and validation set by 80%\20%, stratified wrt. labels. The splits are shown in table 1:

Data	Total number	REAL / FAKE
Training	80,000	40,000 / 40,000
Validation	20,000	10,000 / 10,000
Test	20,000	10,000 / 10,000

Table 1. Overview of the CIFAKE dataset.

For each pixel in the images and for each color channel, the values are normalized to range $[0, 1]$ by using the mean (μ) and standard deviation (σ) across all the training images:

$$\text{normalized pixel value} = \frac{\text{pixel value} - \mu}{\sigma} \quad (1)$$

When loading the training data a batch size = 64 is used and the data is shuffled to enhance generalization, while loading the validation and test data a batch size = 64 is used without shuffling for consistent evaluation.

2.2. Reproducibility

To ensure reproducibility, `torch.manual.seed(409)`, `np.random.seed(409)` and `random.seed(409)` have been set for random number generation. As the notebook has been executed using the T4 GPU with high RAM enabled in Google Colab, we have enforced deterministic behavior by

setting `os.environ["CUBLAS_WORKSPACE_CONFIG"] = ":4096:8"`, and by using `torch.backends.cudnn` to set `deterministic=True` and `benchmark=False`. This procedure works as intended; however Colab adjusts hardware availability depending on the overall demand from users [6], which causes the results to vary slightly, when the code is executed on different days. As we have been refactoring the code since the poster presentation to improve the explainability of the code, how the results are displayed and the quality of plots, we have chosen to include the most recent results in this article. The results from the poster and brief descriptions of how the plots have been improved can be found in the appendices A, B, C and D. The notebooks for each model can be accessed in this GitHub repository.

2.3. Network Architecture

The network architecture has been determined during a trial-and-error process, in which increasing and decreasing the number of convolutional and fully connected layers has been tried, while tuning various hyper-parameters. The network contains 4 convolutional blocks, where the increasing number of channels at deeper layers allows the network to learn more complex and abstract features. The first 3 blocks contains a convolutional layer followed by a ReLU activation function, max-pooling, batch normalization, and a drop-out = 0.3, while the fourth block is followed by a global average pooling instead of max-pooling, which aggregates spatial information into a single value per feature map channel before the fully connected layers. The first fully connected layer is followed by a ReLU activation function, batch normalization, and a drop-out = 0.5, while the output layer is a fully connected layer that depends on which loss function is applied. When applying cross-entropy, the loss function combines log-softmax and negative log-likelihood and the output dimension is 2. When applying binary cross-entropy, the output dimension is 1 and the layer is followed by a sigmoid activation function for probabilistic output modeling. The architecture is shown in figure 1:

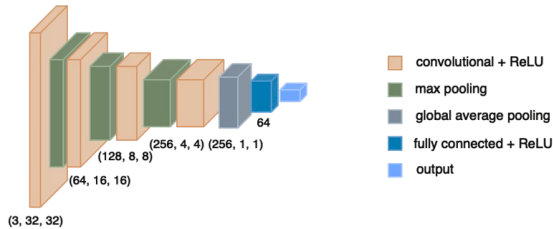


Fig. 1. The overall CNN architecture.

For better convergence, appropriate weight initialization has been applied to the layers, such the `kaiming_normal_` is applied to the convolutional layers and `xavier_normal_` is applied to the fully connected layers.

2.4. Training and Evaluation

As mentioned in section 2.3, we compare the performance when applying the cross-entropy and the binary cross-entropy loss functions, while the rest of the hyper-parameter are fixed. The network is trained using the AdamW optimizer with weight decay $L_2 = 10^{-4}$. The justification for using AdamW instead of the commonly used Adam, is that it applies the weight decay directly to the weights during the update step, instead of adding it to the gradients, which was reflected in the models' performance. The learning rate has been initialized to 10^{-3} and a learning rate scheduler `ReduceLROnPlateau` is applied, such that it reduces the learning rate by factor 0.1 when the validation loss has not decreased during 3 epochs; however the learning rate can not decrease below 10^{-8} . Furthermore, the number of epochs is initialized to 100, but to prevent overfitting, early stopping with patience = 15 is implemented.

In order to evaluate the performance of the models, we use following metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

where TP is the true positive rate, TN is the true negative rate, FP is the false positive rate and FN is the false negative rate.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

2.5. GradCAM

GradCAM (Gradient-weighted Class Activation Mapping) is a technique designed to make CNN models interpretable by visualizing the regions of an image that are most important for a given prediction [7]. It operates by computing the gradients of a target class with respect to the feature maps of the final convolutional layer. These gradients are then pooled to obtain the importance of each feature map channel. GradCAM overlays this information as a heatmap on the original image, providing a visual cue for the areas influencing the model's decision. This interpretability is crucial in practical applications such as forensic analysis and media verification, where understanding model decisions can enhance trust and ensure accountability.

3. RESULTS AND ANALYSIS

3.1. Training

The training and validation loss and accuracy curves are shown in figure 2. For the cross-entropy model, the training

loss decreased consistently, while the validation loss fluctuated after an initial decline. The training accuracy approached nearly 97.5%, but the validation accuracy leveled off at nearly 95%. For the binary cross-entropy model, the training loss decreased and achieved a low value, while the validation loss initially decreased, it generally fluctuated more. The training accuracy approached nearly 97%, but the validation accuracy leveled off at nearly 94%.

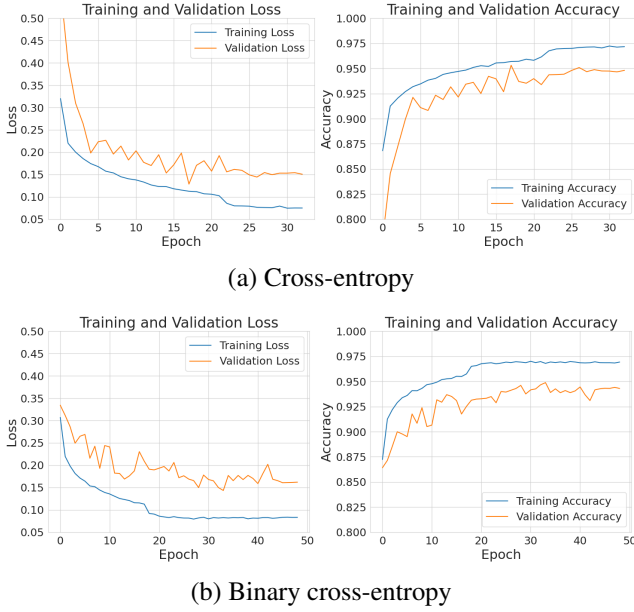


Fig. 2. Loss and accuracy curves for training and validation.

Although both models achieved high training accuracies, the model trained with cross-entropy converged faster, while demonstrating superior validation performance, with reduced fluctuations in validation loss and a closer match between training and validation accuracy.

3.2. Performance Metrics

The evaluation of the two models is summarized in table 2. The cross-entropy model achieved the highest accuracy of 95.2 %, with precision, recall, and F1-score all exceeding 95 %. Binary cross-entropy performed slightly lower, with an accuracy of 94.7 %, precision of 95.1 %, and recall and F1-scores of 94.7 %.

Loss function	Accuracy	Precision	Recall	F1-score
Cross-entropy	95.2	95.4	95.2	95.1
Binary cross-entropy	94.7	95.1	94.7	94.7

Table 2. Performance metrics for evaluation of the quality of the models' performance, where the values are in %.

3.3. Confusion Matrices

The confusion matrices, illustrated in figure 3, highlight the classification performance of the models for both real and synthetic images. For the cross-entropy loss function, the model accurately classified 49.47% of the images as real and 45.68% as synthetic. Misclassifications include 0.53% real images identified as synthetic and 4.32% synthetic images identified as real. Similarly, the binary cross-entropy model accurately classified 49.45% correct classifications for real images and 45.28% for synthetic ones. The binary cross-entropy model reduced the misclassification rates slightly, with 4.71% synthetic images identified as real and 0.55% real images identified as synthetic.

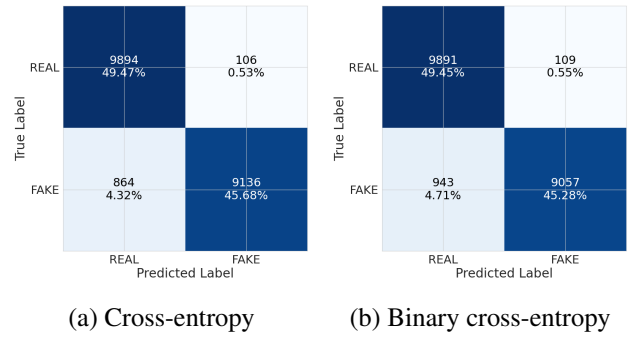


Fig. 3. Confusion matrices for evaluation of the models' performance.

3.4. GradCAM Analysis

GradCAM has been applied to highlight regions of interest that the models used to classify images as real or synthetic. The visualizations of the models shown in figure 4 revealed:

1. True label = REAL, Predicted label = REAL:

- Both models highlight regions in the image that the CNN considers most relevant to its prediction.
- For both models the focus appears to be on parts of the image containing textures or patterns indicative of real-world details, such as the horse's mane and background textures.
- The overlays demonstrates alignments between the highlighted regions and features expected in real images.

2. True label = FAKE, Predicted label = FAKE:

- Both models highlights artifacts or features in the image that might signal artificial generation, such as abrupt transitions, pixel patterns, or regions that lack natural randomness.
- The overlays confirms that the highlighted areas correspond to synthetic artifacts.

3. True label = REAL, Predicted label = FAKE:

- Both models illustrates a failure in prediction. The heatmap shows the areas the CNN considered influential in making its (incorrect) prediction.
- The highlighted regions might correspond to patterns the model misinterpreted as synthetic artifacts, such as reflective areas or smooth patterns.
- The overlays indicate that the model struggles with certain natural patterns that superficially resemble synthetic features.

4. True label = FAKE, Predicted label = REAL:

- Here, both models incorrectly classified a synthetic image as real. The heatmap focuses on regions that the model incorrectly deemed "natural".
- The overlays show that these regions mimic real-world features closely enough to confuse the model.

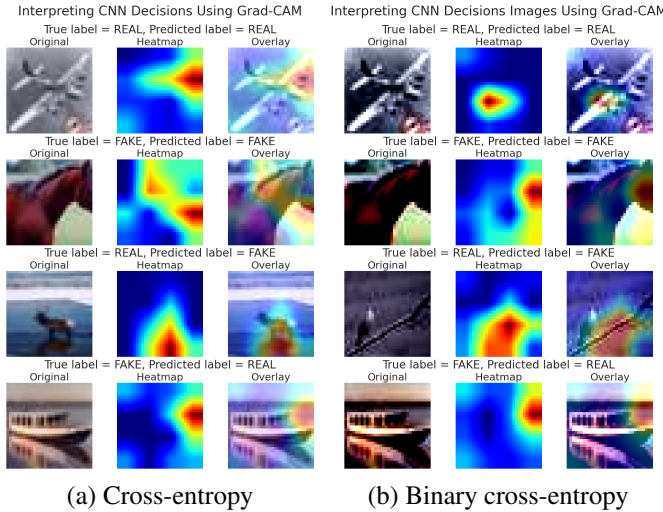


Fig. 4. Comparison of GradCAM visualizations for both models. The highlighted regions indicate areas of focus contributing to the classification outcome.

4. DISCUSSION AND FUTURE WORK

The GradCAM visualizations, paired with the confusion matrices and training-validation curves, provide key insights into the strengths and limitations of the CNN models. Both models achieved high training accuracy, with the cross-entropy loss function slightly outperforming binary cross-entropy in terms of validation accuracy and stability. This is reflected in the smoother and more consistent convergence of the cross-entropy model's training and validation loss curves, suggesting better generalization. Conversely, the binary

cross-entropy model showed greater fluctuations in validation loss, indicative of potential sensitivity to certain data features or overfitting tendencies in some epochs.

Both models exhibit high performance in correctly classifying REAL and FAKE images, as indicated by the dominant diagonal entries in the confusion matrices. GradCAM visualizations confirm that the models focus on meaningful regions in these cases, such as textures or artifacts characteristic of real or synthetic images. This indicates that the models effectively learn and leverage discriminative features for their predictions. However, both models struggle with misclassifications, particularly in cases involving reflective artifacts. These reflections challenge the models' ability to distinguish real-world textures from synthetic imitations, as reflections can exhibit similar patterns in both real and synthetic images, such as smooth gradients, specular highlights, or ambiguous light sources, leading to confusion during classification. For false negatives, the GradCAM heatmaps show that the models focus on ambiguous regions with insufficient texture, leading to misinterpretation as synthetic. In contrast, for false positives, the models are misled by synthetic images with reflections that closely mimic real-world features.

The confusion matrices further highlight differences in misclassification rates between the models. The binary cross-entropy model demonstrates fewer false negatives, suggesting better recall for FAKE images, but at the cost of slightly higher false positives. This indicates improved sensitivity to synthetic patterns, likely due to broader attention observed in the heatmaps. The cross-entropy model, on the other hand, exhibits fewer false positives, reflecting a tighter focus on specific regions and a tendency to prioritize precision over recall. These findings suggest that the choice of loss function influences model behavior: binary cross-entropy is better suited for applications prioritizing the detection of FAKE images, while cross-entropy provides a more balanced approach between precision and recall that is reflected by the higher F1-score.

These findings highlight opportunities for further enhancement. While the models perform very well, incorporating data augmentation strategies with e.g. more reflective artifacts could help address specific challenges, such as distinguishing real-world textures from synthetic reflections. Additionally, testing the models on higher-resolution datasets with more complex textures would help evaluate their scalability to real-world applications, such as forensic analysis or media verification.

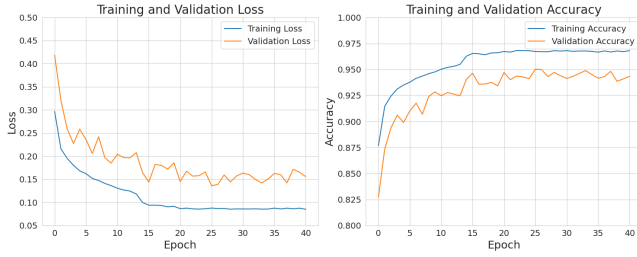
The training and validation curves further underscore the importance of appropriate regularization and early stopping. Although both models incorporated dropout and learning rate schedules to mitigate overfitting, the higher validation loss variance by the binary cross-entropy model suggests room for improvement in hyper-parameter tuning. Exploring advanced optimization techniques, such as cyclical learning rates or fine-grained weight decay, could further stabilize training.

5. REFERENCES

- [1] Jordan J. Bird and Ahmad Lotfi, “Cifake: Image classification and explainable identification of ai-generated synthetic images,” *arXiv*, vol. abs/2303.14126, 2023.
- [2] Md. Zahid Hossain, Farhad Uz Zaman, and Md. Rakibul Islam, “Advancing ai-generated image detection: Enhanced accuracy through cnn and vision transformer models with explainable ai insights,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*. IEEE, 2023.
- [3] A. Lokner Laević, T. Kramberger, R. Kramberger, and D. Vlahsek, “Detection of ai-generated synthetic images with a lightweight cnn,” *AI*, vol. 5, no. 3, pp. 1575–1593, 2024.
- [4] G. Bharathi Mohan, P. K. Rangarajan, S. Akilesh Rao, M. Suresh, M. Abinandhini, and Y. Jaikanth, “Detecting ai-generated images with cnn and interpretation using explainable ai,” in *2024 IEEE International Conference on Contemporary Computing and Communications (InC4)*. IEEE, 2024.
- [5] Jordan J. Bird, “Cifake: Real and ai-generated synthetic images,” <https://www.kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images>, 2023, Accessed: 2024-11-04.
- [6] Google, “Frequently asked questions,” <https://research.google.com/colaboratory/faq.html#gpu-availability>, 2024, Accessed: 2024-12-20.
- [7] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra, “Grad-cam: Visual explanations from deep networks via gradient-based localization,” *International Journal of Computer Vision*, vol. 128, no. 2, pp. 336–359, 2020.

A. APPENDIX

Due to the limited reproducibility explained in section 2.2 caused by executing the code in Google Colab, we have different loss and accuracy curves on the poster than in this article, as we have chosen to include our most recent results. However, it is noticeable that we observe the same trend of the curves converging faster when applying the cross-entropy loss function than when applying the binary cross-entropy in figure 5. However, the loss and accuracy curves for the cross-entropy model presented in figure 2 shows smoother loss and accuracy for the validation set when converging and lower training loss.



(a) Cross-entropy



(b) Binary cross-entropy

Fig. 5. Loss and accuracy curves for training and validation.

For the plots shown in figure 2, we increased all font sizes and moved the legends in the accuracy curves to the lower right corner to make them easier to read and understand.

B. APPENDIX

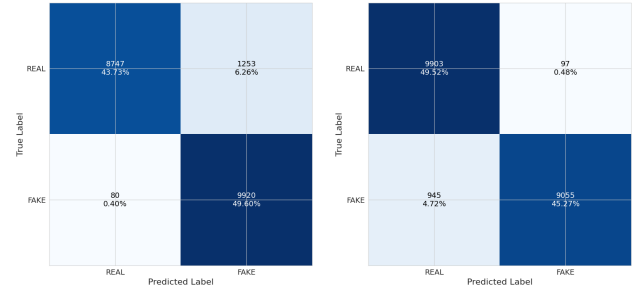
Keeping the observations described in appendix A in mind, we presented different values for the performance metrics on the poster than in this article. This is may be caused by the cross-entropy model in figure 2 appearing more stable overall, with better-aligned training and validation metrics, indicating stronger generalization. While the binary cross-entropy model in figure 5 shows slower convergence and higher variability in validation performance.

Loss function	Accuracy	Precision	Recall	F1-score
Cross-entropy	93.3	93.9	93.3	93.3
Binary cross-entropy	94.8	98.9	90.5	94.6

Table 3. Performance metrics for evaluation of the quality of the models' performance, where the values are in %.

C. APPENDIX

As the confusion matrices in figure 6 illustrate the classification performance in table 3 of the models, they are different from the confusion matrices in figure 3. However, the confusion matrices in figure 3 reflect the increase in F1-score for both models.



(a) Cross-entropy

(b) Binary cross-entropy

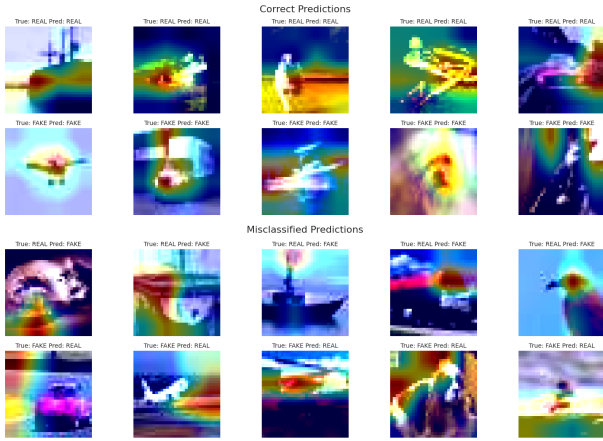
Fig. 6. Confusion matrices for evaluation of the models' performance.

For the plots shown in figure 3, we increased all font sizes to make them easier to read and understand.

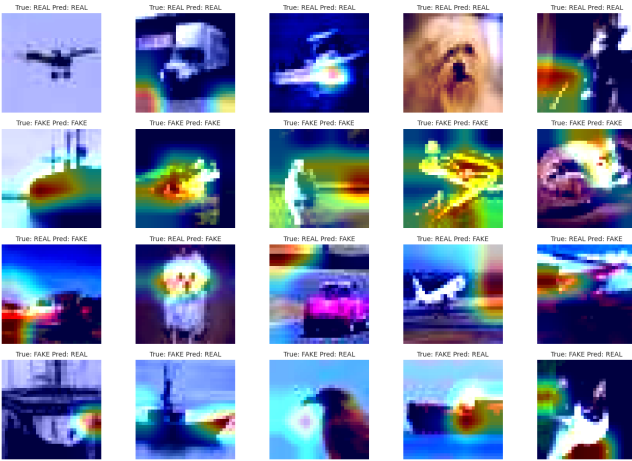
D. APPENDIX

For the poster presentation, we were inspired by the article "Advancing AI-Generated Image Detection: Enhanced Accuracy through CNN and Vision Transformer Models with Explainable AI Insights" [2] to generate the GradCAM overlay on 5 samples from the test set for correctly and misclassified images, where "True" is the true label and "Pred" is the predicted label. The correct predictions are shown in the 2 rows at the top, and the misclassified predictions are shown in 2 rows at the bottom, which is displayed as subtitles in figure (a). However, we missed displaying these subtitles in figure (b).

However, as mentioned in section 2.2, we have been refactoring the code after the poster presentation. We have chosen to display one sample from the test set for each model instead in figure 4, where we present the original images, then the GradCAM filter and then the overlay for the correct and misclassified images to ease the interpretability.



(a) Cross-entropy



(b) Binary cross-entropy

Fig. 7. Images with GradCAM overlay.

When inspecting the notebooks for both models, the code for these plots is shown at the bottom along with some code for displaying these images in the same format without the GradCAM overlay.